

RESEARCH ARTICLE

Kaempferia parviflora extract ameliorates the cognitive impairments and the reduction in cell proliferation induced by valproic acid treatment in rats



Jariya Umka Welbat^{a,b,c,*}, Pornthip Chaisawang^a, Wunnee Chaijaroonkhanarak^a, Parichat Prachaney^a, Wanassanun Pannangrong^{a,c}, Bungorn Sripanidkulchai^c, Peter Wigmore^d

^a Department of Anatomy, Faculty of Medicine, Khon Kaen University, Khon Kaen, 40002, Thailand

^b Neuroscience Research and Development Group, Khon Kaen University, Khon Kaen, 40002, Thailand

^c Center for Research and Development of Herbal Health Products, Khon Kaen University, Khon Kaen, 40002, Thailand

^d School of Life Sciences, Medical School, Queen's Medical Centre, The University of Nottingham, Nottingham, NG7 2UH, UK

ARTICLE INFO

Article history:

Received 28 October 2015

Received in revised form 11 April 2016

Accepted 15 April 2016

Keywords:

Kaempferia parviflora

Valproic acid

Hippocampus

Neurogenesis

Spatial memory

ABSTRACT

Kaempferia parviflora is a herbal plant whose rhizomes are used in traditional medicine. Investigations of this plant have shown it to have antidepressant activity and to improve learning and memory in animal models. The aim of the present investigation was to determine whether *K. parviflora* could protect the brain from the impairments in cognition and hippocampal neurogenesis which are caused by valproic acid (VPA).

Male Sprague Dawley rats (180–200 g) were given once daily *K. parviflora* extract (100 mg/kg) via oral gavage for 21 days. Rats received twice daily intraperitoneal injections of valproic acid (300 mg/kg) from days 8 to 21 of the experiment. Spatial memory was tested using the novel object location (NOL) test five days after the end of treatment. Cell proliferation in the sub granular zone (SGZ) of the dentate gyrus was quantified by immunohistochemistry and levels of doublecortin (DCX) were determined by Western blotting.

Co-treatment of VPA and *K. parviflora* prevented the cognitive decline and reduction in proliferating cells caused by VPA. Furthermore, co-treatment significantly increased DCX protein levels within the hippocampus.

These findings demonstrate that *K. parviflora* is able to prevent the brain from VPA-induced the impairments of spatial memory and proliferating cells within the SGZ.

© 2016 Elsevier GmbH. All rights reserved.

1. Introduction

Valproic acid (VPA) is clinically used as a long-term anti-epileptic drug to prevent seizures in both adults and children (Buckley, 2008). It is thought that VPA acts by blocking sodium and calcium channels to reduce neuronal activity. Consequently, GABA and aspartate levels within the brain are elevated and reduced, respectively (Kwan et al., 2001). VPA also functions as a histone deacetylase inhibitor by affecting chromatin modification, which results in a decrease in cell proliferation and an increase in neuronal

differentiation in the cells involved in neurogenesis (Hsieh et al., 2004; Kostrouchova et al., 2007). VPA treatment reduces cell proliferation within the sub granular zone (SGZ) of the dentate gyrus (DG) and so inhibits hippocampal neurogenesis (Umka et al., 2010). Several studies have reported that patients treated with VPA develop mild to moderate cognitive deficits (Hommet et al., 2007; Mula and Trimble, 2009). In an animal model, VPA has been shown to reduce performance in a spatial memory test and this behavioral change was associated with a reduction in cell proliferation required for hippocampal neurogenesis (Umka et al., 2010). Although, VPA has been generally considered to have a good safety profile, the possibility of drug-induced cognitive dysfunction must be considered in all patients treated with this drug.

Adult neurogenesis, the process by which neural stem cells continuously generate new neurons throughout life, has been detected in all mammals, including the brains of rodents and humans

* Corresponding author at: Department of Anatomy, Faculty of Medicine, Khon Kaen University, Khon Kaen 40002, Thailand. Tel.: +66 846 976 105; fax: +66 43 348 381.

E-mail address: jariya@kku.ac.th (J.U. Welbat).

(Ehninger and Kempermann, 2008; Eriksson et al., 1998). The SGZ of the hippocampal dentate gyrus is one of two main regions where adult neurogenesis takes place. In this location, neural stem and progenitor cells proliferate to generate new dentate granule neurons which are incorporated into the circuitry of the hippocampus (Abrous et al., 2005; Ehninger and Kempermann, 2008). In rodents, approximately 9000 new cells are produced each day and this process is required for hippocampal dependent learning (Cameron and McKay, 2001). As neural progenitors differentiate into mature neurons, they transiently express the microtubule associated protein doublecortin (DCX), levels of which provide an additional measure of the rate neurogenesis (Couillard-Despres et al., 2005). Brain-derived neurotrophic factor (BDNF), a polypeptide growth factor, is produced from granule cells in the hippocampal dentate gyrus (Conner et al., 1997; Wetmore et al., 1990) and functions to up-regulate neuronal cell proliferation and survival (Linnarsson et al., 2000). Expression of BDNF Levels is associated with learning and adult neurogenesis (Nibuya et al., 1995; Shirayama et al., 2002). Antidepressants increase the level of BDNF expression that leads to induction of adult hippocampal neurogenesis (Duman et al., 2001). Adult hippocampal neurogenesis is thought to provide additional plasticity within the hippocampus, which is required in the process of memory consolidation (Ramirez-Amaya et al., 2006). Neurogenesis in the hippocampus can be influenced by many factors, including drugs and environmental stimuli (Kempermann, 2006). It has recently been shown that antidepressants play a significant role as a modulator of neurogenesis in the adult hippocampus (Duman et al., 2001). Chronic (>2 weeks) antidepressant administration promotes adult hippocampal neurogenesis, a response which may be essential for the mood altering and cognitive effects of these drugs (Santarelli et al., 2003). For example, antidepressants counteract both the decrease in hippocampal neurogenesis and the behavioral changes caused by stress (Kodama et al., 2004) and chemotherapeutic drugs (Lyons et al., 2012).

Kaempferia parviflora, locally called Krachai Dam, is a plant originally found in the North and Northeast of Thailand and is classified in the family Zingiberaceae. Rhizomes of *K. parviflora* are utilized as a traditional medicine for a variety of illnesses, including peptic ulcers and inflammation. Extracts have also been shown to have anti-mutagenic, antibacterial, anti-fungal, anti-allergic and anti-viral properties (Mekjaruskul et al., 2012). *K. parviflora* at a dose of 100 mg/kg has been shown to have antidepressant-like effects (Wattanathorn et al., 2007). If *K. parviflora* increases cell proliferation, and this is correlated with improved spatial memory, this would suggest that it is acting in the same way as conventional antidepressants (Surget et al., 2008). It is, however, unknown whether *K. parviflora* increases hippocampal neurogenesis and cognition in the same way as other antidepressants. Therefore, the first objective of this study was to investigate the effect of *K. parviflora* on cell proliferation and neuronal differentiation in the SGZ of the hippocampal dentate gyrus and whether this was correlated with changes in performance in a hippocampus-dependent spatial memory test in rats. Following this, the plant extract was assessed for its ability to ameliorate the impact of VPA on cell proliferation in the SGZ and spatial memory.

2. Materials and methods

2.1. Plant material and extraction

K. parviflora rhizomes were obtained from Loei province, Thailand and authenticated. The voucher specimen was deposited at the Center for Research and Development of Herbal Health Products, Khon Kaen University, Thailand. The dried plant powder was extracted by maceration in 95% ethanol as previously

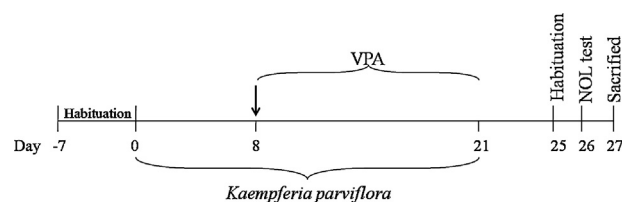


Fig. 1. Timeline of drug administration and behavioral testing. Brackets show the period that rats received VPA and *K. parviflora*. Animals were put down one day after the Novel object location (NOL) test.

reported (Mekjaruskul et al., 2012). As detected by HPLC method, the extract contained three major methoxyflavones that are 3,5,7,3',4'-pentamethoxyflavone, 5,7,4'-trimethoxyflavone and 5,7-dimethoxyflavone at concentration of 23.32, 31.06 and 21.10 mg/g, respectively.

2.2. Animals

Male Sprague Dawley rats (the National Laboratory Animal Center, Mahidol University, Salaya, Nakornpathom) weighing 180–200 g were housed in cages of 3 or 4 and maintained with 12 h light/dark cycle (7:00/19:00 h). Food and water were provided *ad libitum*. Principles of laboratory animal care in this animal study were in accordance with the Ethics of the Animal Experiment of National Research Council of Thailand and with approval from the Animal Ethics Committee of Khon Kaen University under permit number AEKKU 49/2556.

2.3. Drug treatment protocols

Rats were weighed daily from arrival and allowed to habituate for 1 week prior to drug administration. Forty rats were randomly allocated to vehicle, *K. parviflora*, fluoxetine, VPA and *K. parviflora* plus VPA groups. The vehicle group received 0.5% carboxymethylcellulose (Fluka, USA) by gavage for 21 days and received 0.9% sterile saline by intraperitoneal (i.p.) injection twice daily for 14 days from days 8 to 21. The *K. parviflora* groups received *K. parviflora* (100 mg/kg, dissolved in 0.5% carboxymethylcellulose) by gavage for 21 days. The VPA groups were given two daily i.p. injections of VPA (300 mg/kg dissolved in 0.9% saline; Sigma–Aldrich, USA) for 14 days from days 8 to 21 (Fig. 1). This VPA treatment procedure significantly decreases seizure incidence in epileptic rats (Nissinen and Pitkanen, 2007).

2.4. Behavioral analysis (novel object location (NOL) spatial memory task)

Five days after drug or vehicle administration, rats were tested using the NOL test, adapted from Dix and Aggleton (1999). The experiments were conducted at an illumination of 350–400 lx between 09:00 and 16:00 h. The procedure consisted of habituating the animals for 30 min to the arena (36 cm × 50 cm × 30 cm) 24 h prior to testing and for a further 3 min, 5 min prior to familiarization trial. During the familiarization trial, two objects were randomly placed in separate locations in the arena and the rats were allowed to explore for 3 min. Then, the rats were returned to their home cages for a 15 min (inter-trial interval) during which the arena and the object were cleaned with 20% ethanol. In the choice trial, the rats were returned to the arena for 3 min where one object was returned to the same location while the other was moved to a new position (novel location). Exploration was recorded with a video camera mounted directly above each test arena. Exploration was defined as when the rat sniffed, licked, chewed or directed their nose at a distance less than 2 cm from the object (Dix and Aggleton,

1999). The quantification of exploratory activity was scored twice by an observer blind to the treatment of each group and averaged using a stopwatch from digitized recordings. The preference index (PI) is defined as a percentage of exploration time spent on object in the novel location over the total exploration times of novel and familiar locations in the choice trial as described by Lyons et al. (2012). A PI which is significantly different from chance shows that the animals were spending more time on the object in the novel location and were able to recall the previous object locations.

2.5. Brain tissue preparation

Animals were killed by rapid stunning and cervical dislocation 27 days after treatment and brains were removed. One half of each brain was used for immunohistochemistry and the hippocampal formation dissected out from the other half and snap-frozen for use in Western blotting. For immunohistochemistry, half brains were cryoprotected in 30% sucrose for 3 h at 4 °C before embedding in OCT (Thermo Fisher Scientific, Germany). Tissue was then snap-frozen in liquid nitrogen cooled isopentane and stored at –80 °C prior sectioning. Frozen brains were serially sectioned through the entire dentate gyrus at a thickness of 20 µm in the coronal plane using a cryostat. Sections were thaw mounted on APES coated slides and stored at –20 °C until used for immunohistochemistry.

2.6. Immunohistochemistry

Every 15th section from the length of dentate gyrus was used to select a total of 9 sections per brain using a systemic random sampling technique (Mayhew and Burton, 1988). All reactions were carried out at room temperature in a light-proof humidity chamber. Ki-67 staining was performed as described (Umka et al., 2010). Briefly, sections were incubated with monoclonal mouse anti Ki-67 primary antibody (1:150, Vector Laboratory, Inc., USA.) for 1 h. Sections were then incubated with 488 Rabbit Anti-mouse IgG (1:300, Invitrogen, USA.), followed by 30 sec counterstaining with propidium iodide (1:6000, Sigma–Aldrich, USA).

All sections were viewed at X40 on Nikon ECLIPSE 80i fluorescence microscope. The Ki-67 positive cells within the SGZ, defined as 3 cell diameters from the inside of the dentate gyrus, were counted for both blades of the dentate gyrus. An estimate of the total number of Ki-67 positive cells was produced by multiplying the averaging positive cell counts per section by 15 (Umka et al., 2010).

2.7. Western immunoblotting

Hippocampal tissue was prepared for Western blotting as previously described (Umka et al., 2010). Samples were loaded on to 10% SDS-polyacrylamide gels and proteins were transferred to nitrocellulose membranes. The nitrocellulose membranes were incubated overnight with the following primary antibodies: polyclonal goat anti-DCX C-18 (1:1000; Santa Cruz Biotechnology, USA) and polyclonal rabbit anti-BDNF N-20 (1:200; Santa Cruz Biotechnology, USA). Membranes were incubated with polyclonal goat anti-goat horseradish peroxidase-conjugated secondary antibody (1:2000; DAKO Cytomation, UK) and polyclonal goat anti-rabbit horseradish peroxidase-conjugated secondary antibody (1:2000; DAKO Cytomation, UK) for 1 h at room temperature. Detection of protein bands on membranes were carried out using the chemiluminescence kit (Amersham, GE Healthcare) and recorded using ImageQuant 400 (GE Healthcare, UK). The level of GAPDH (Abcam, UK) was used as a loading control. Quantitative measurement of protein levels was performed using ImageJ software (Wang et al., 2003) by measuring the optical density of each protein band.

2.8. Statistical analysis

All statistical parameters were calculated using GraphPad Prism 5.0 software and significance was regarded as $P < 0.05$. Data were analyzed via paired Student's *t*-test and ANOVA (analysis of variance) test where appropriate.

3. Results

3.1. Behavioral effects of *K. parviflora*

The effect of VPA and *K. parviflora* on spatial working memory was determined using the NOL test. There was no significant difference between groups in both trials for either total exploration time or mean locomotor velocity (Fig. 2c), indicating none of the groups were impaired in their activity. Animals from all groups showed no significant difference in the total exploration time of the two objects in the familiarization trial (Fig. 2a), indicating that the animals did not show a preference for the location of either object. After the 15 min inter-trial interval animals carried out the choice trial. In control, *K. parviflora* and VPA plus *K. parviflora* groups, animals spent significantly more time exploring the object in the novel location when compared to the object in the familiar location ($P < 0.05$, Fig. 2b). This indicates an ability to discriminate between objects in new and familiar locations and a normal level of spatial working memory found in these rodents. In contrast, VPA-treated animals spent similar amounts of time exploring objects in both novel and familiar locations ($P < 0.05$, Fig. 2b). This result shows that VPA treatment caused a spatial working memory deficit. For further analysis, object exploratory times in the choice trial were converted into a preference index (PI) to allow for between-group comparison. A one-sample *t* test showed that the mean PI of the control, *K. parviflora* and *K. parviflora* plus VPA groups was significantly higher than 50% chance ($P < 0.05$, Fig. 2d). However, the PI of the VPA group was not significantly different from chance. Using the PI to compare between groups, showed that VPA-treated group was significantly different from the other two groups ($P < 0.05$). These results indicate that VPA treatment causes spatial memory deficits which are improved by co-administration with *K. parviflora*.

3.2. Effects of treatments on cell proliferation in the SGZ of the dentate gyrus

Cell proliferation was quantified by counting Ki-67 positive cells in the SGZ of the DG (Fig. 3a–d). The total number of Ki-67 positive cells was significantly different among groups ($F = 12.76$, $P < 0.0001$, one-way repeated measures ANOVA). Bonferroni *post hoc* tests showed that the number of Ki-67 positive cells in VPA treated animals was significantly decreased by 45% in comparison with controls ($P < 0.05$). Treatment with *K. parviflora* on its own did not alter Ki-67 positive cell numbers compared to controls ($P > 0.05$), however, receiving *K. parviflora* alone significantly increased Ki-67 positive cell numbers compared to VPA ($P < 0.05$). Co-administered of *K. parviflora* and VPA showed no significant difference from control animals but was significantly higher than the VPA alone group ($P < 0.05$). These results indicate that *K. parviflora* can prevent the decrease in progenitor cell division found after VPA treatment.

3.3. Effects of treatments on the expression of DCX and BDNF in hippocampus

The expression of DCX and BDNF in the hippocampus at the end of the experiment was determined by Western blotting (Fig. 4). There was no significant difference of DCX levels when compared between groups ($F = 2.523$, $P = 0.060$, one-way repeated measures ANOVA, Fig. 4a). Animals receiving *K. parviflora* alone showed no

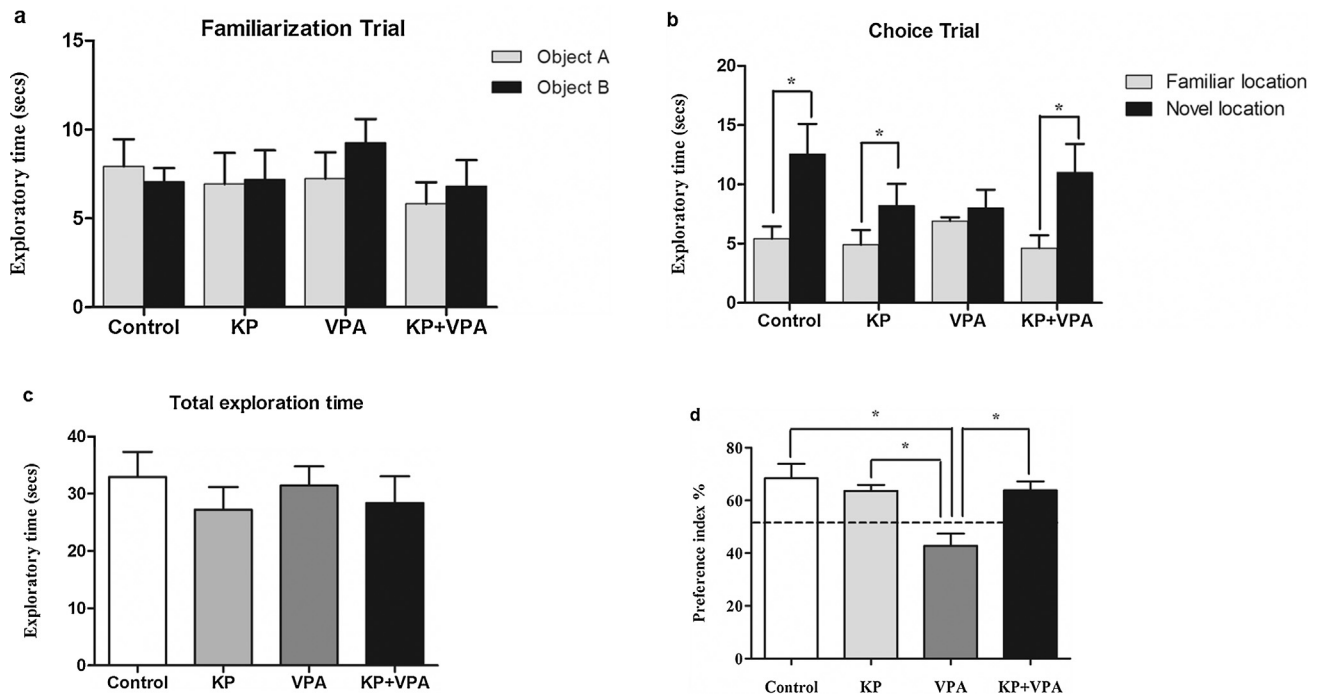


Fig. 2. Mean exploration times (mean $\pm$ SEM) of animals for each object in the familiarization (a) and choice (b) trials of the NOL test. In the familiarization trial, none of the groups spent significantly different amount of time exploring either of the two objects. In the choice trial, control, fluoxetine, *Kaempferia parviflora* (KP) and KP plus VPA animals spent more time exploring the object in the new location ($*P < 0.05$) but VPA-treated animals did not show a significant difference between the two locations. The combined total exploration time of familiarization and choice trials (c) did not differ significantly between groups. The Preference index (PI) of the control, fluoxetine, KP and KP plus VPA groups was significantly higher compared 50% chance ($*P < 0.05$, d). Additionally, the PI of the VPA group was significantly different when compared with the other three groups ($*P < 0.05$, d).

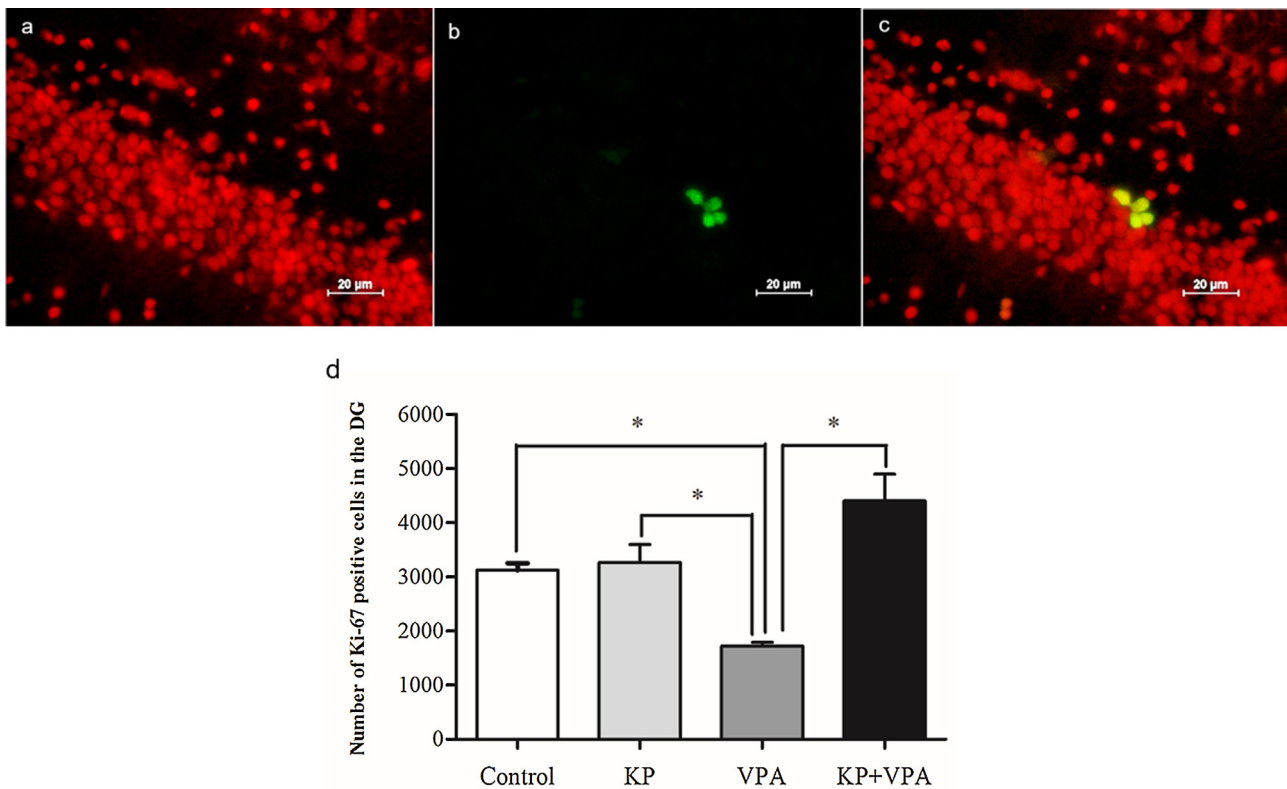


Fig. 3. Photos showing positive immunostaining for the proliferating cell marker Ki-67 (green, b), nuclear stain propidium iodide (red, a) and merged (c). Mean Ki-67 positive cell counts of VPA-treated group were significantly lower than control, fluoxetine, *K. parviflora* (KP) and KP plus VPA groups ($*P < 0.05$, d).

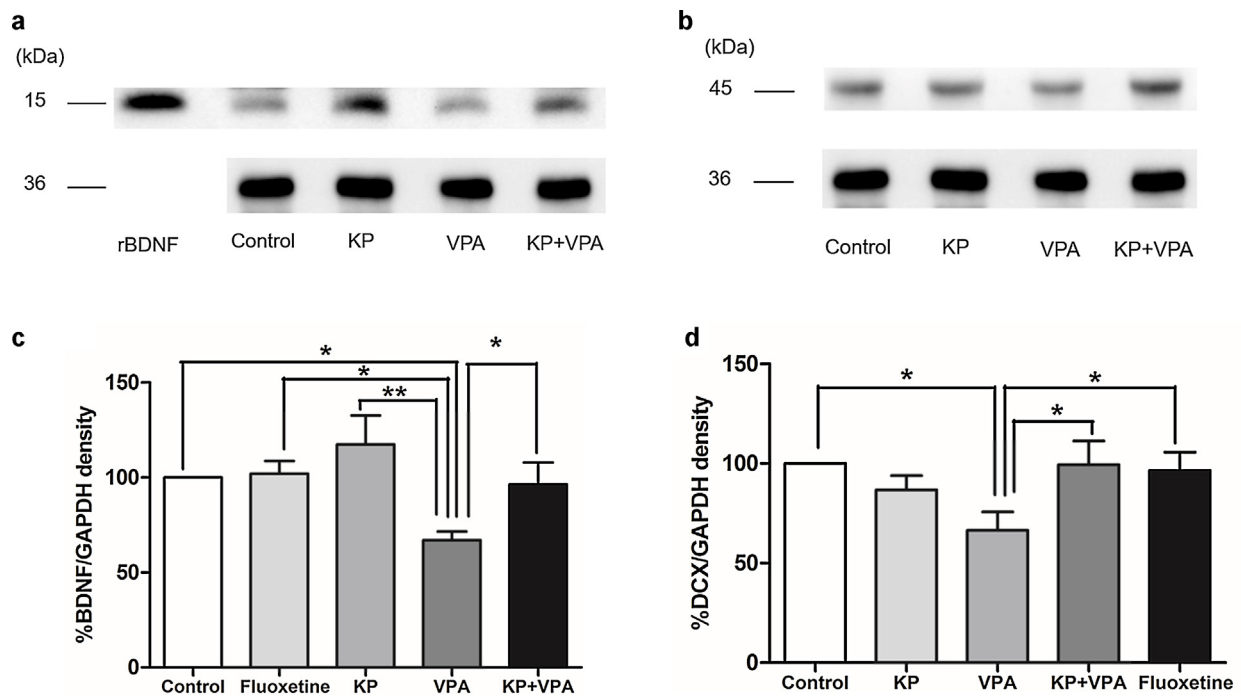


Fig. 4. Determination of DCX and BDNF protein levels in the hippocampus. Levels of DCX in animals receiving VPA were significantly lower than animals in control, fluoxetine, *K. parviflora* (KP) or both KP and VPA groups ($P < 0.05$, a and b). VPA itself also significantly decreased BDNF levels when compared with control, fluoxetine, or both KP and VPA ($* P < 0.05$; $** P < 0.01$, c and d), whereas VPA group was not different from KP group.

significant difference in DCX levels compared to controls ($P > 0.05$). Animals receiving VPA on its own showed a significant reduction in DCX levels when compared to control levels ($P = 0.017$). DCX levels of animals receiving both VPA and *K. parviflora* showed significantly higher than animals treated with VPA alone ($P = 0.012$) but not significantly different from control group ($P = 0.966$ and 0.812 , respectively). These findings suggest that co-administration of *K. parviflora* and VPA increases DCX protein expression above levels found with VPA on its own. The comparison of BDNF protein expressions showed that there was a significant difference between groups ($F = 3.879$, $P = 0.014$, one-way repeated measures ANOVA, Fig. 4b). BDNF levels in VPA-treated animals were significantly lower than control group ($P = 0.019$ and 0.014 , respectively). Animals receiving VPA had BDNF levels less than *K. parviflora* animals ($P = 0.001$), however, BDNF levels of VPA and *K. parviflora* co-treatment animals were found to be significantly higher than VPA treated animals ($P = 0.036$). These results indicate that *K. parviflora* can ameliorate the impact of VPA on BDNF levels in the hippocampus.

4. Discussion

In the present study, we investigated the possible protective properties of *K. parviflora* in preventing a cognitive deficit and decreased cell proliferation in the SGZ of the hippocampal dentate gyrus found after chronic VPA treatment. Co-administration of *K. parviflora* improved the impaired spatial working memory of VPA treated animals and prevented the reduction in cell proliferation within the SGZ.

VPA has previously been shown to impair cognition in both animals (Umka et al., 2010) and human patients (Mula and Trimble, 2009) and this was confirmed in the results presented within our study. The NOL behavioral test is hippocampal dependent and uses the natural preference of rats to explore an object located in a novel position without requiring negative and positive reinforcements (Dix and Aggleton, 1999). In the present study, animals which had received VPA showed impaired discrimination between objects in

novel and familiar locations, indicating a deficit in spatial memory. Treatment with *K. parviflora* alone for 21 days did not improve spatial memory but animals administered with both VPA and *K. parviflora* showed an improvement in their ability to distinguish between novel and familiar location. Comparison of the PI between groups in the choice trial showed that VPA-treated animals showed significantly less discrimination than all other groups. This suggests that *K. parviflora* provides a cognitive benefit to animals subjected to sub-chronic VPA treatment. Previous studies have shown that *K. parviflora* can enhance spatial memory performance and reduce the memory deficits found in stressed rats (Hawiset et al., 2011).

The dentate gyrus of the hippocampus is closely associated with spatial memory performance, a property which requires the additional neural plasticity afforded by hippocampal neurogenesis. The SGZ of the dentate gyrus is a narrow region found on the inside the blades of the dentate gyrus where neural stem cells generate new granule cell neurons. More than 70% of dividing cells in the SGZ differentiate to become neurons in the dentate gyrus (Zhao et al., 2006). Spatial memory performance is strongly affected by levels of hippocampal neurogenesis and for this reason we wished to see if there was any correlation between cognitive changes produced by different treatments and neurogenesis. To determine the effects of *K. parviflora* and VPA treatment on cell proliferation in the SGZ, we quantified the number of Ki-67 positive cells by immunostaining. The results show that the number of Ki-67 positive cells in VPA-treated animals was significantly lower than in control animals, confirming earlier results (Hsieh et al., 2004; Umka et al., 2010). While treatment with *K. parviflora* on its own had no effect on cell proliferation, co-administration of *K. parviflora* with VPA caused levels of cell proliferation to remain at control levels. This indicates a protective action of *K. parviflora*, which was able to prevent the cognitive decline associated with a reduction in hippocampal neurogenesis.

DCX is known to be transiently expressed in proliferating cells during final mitosis and their subsequent differentiation into neurons (Couillard-Despres et al., 2005). The DCX protein is mainly expressed in the immature neurons in the SGZ of the dentate gyrus

between 3 days to 3 weeks, and levels of DCX rises to a maximum within 1 week after division (Winocur et al., 2014; Zhang et al., 2014). DCX marker has been used to assess the proliferation and differentiation of neural progenitor cells (NPCs) in the dentate gyrus (Mukuda et al., 2014; Zhang et al., 2014). VPA caused a significant drop in hippocampus DCX protein levels compared with controls a result in disagreement with our previous study (Umka et al., 2010). The discrepancy of DCX levels in the hippocampus might be due to the duration of VPA treatment. Longer period of VPA treatment (14 days) in this study might cause more impaired immature neurons when compared with our previous study (10 days). Co-treatment of VPA with *K. parviflora*, however, significantly increased DCX levels compared with animals treated with VPA on its own. Taken together, the data in the present study indicate that *K. parviflora* per se does not enhance survival of immature neurons. However, *K. parviflora*-induced increase in DCX levels might be considered to stimulate neurogenesis by increasing survival of newborn neurons and proliferating NPCs in animals treated with VPA.

Alcoholic extract of *K. parviflora* contains numerous flavonoids that have potentially neuroprotective potential and antioxidant activity. Dietary flavonoids have an ability to enhance memory, learning and cognitive function by increasing production of BDNF (Spencer, 2009). Moreover, *K. parviflora* has been shown to exert antidepressant effects (Wattanathorn et al., 2007). There is abundant evidence that antidepressants can improve memory deficits and hippocampal cell proliferation and survival when these are induced by chemotherapy treatment (Lyons et al., 2012). Antidepressants increase the level of BDNF and indirectly induce neurogenesis by directly stimulating generation of hippocampal neural stem cells (Sairanen et al., 2005). Besides affecting neurogenesis, BDNF is known to play a crucial role in the process of memory consolidation (Bekinschtein et al., 2014). Determination of BDNF levels in the hippocampus from the present study reveals that *K. parviflora* itself did not affect BDNF expression. However, BDNF levels in animals receiving both VPA and *K. parviflora* were improved to reach control and fluoxetine levels. These results suggest that *K. parviflora* might play a crucial role to prevent VPA-induced a decrease of cell proliferation, survival and memory by improving levels of both BDNF in the hippocampus.

In conclusion, the results of this study show that co-administration of *K. parviflora* can counteract the cognitive impairments associated with a reduction in the proliferation of neural precursors in the SGZ of the hippocampus caused by VPA. Our results provide evidence that this plant might protect a number of conditions involving cognitive impairment and reduced hippocampal neurogenesis.

Acknowledgements

The authors would like to thank Center for Research and Development of Herbal Health Products, Invitation Research of the Faculty of Medicine (grant no. I 57107), Neuroscience Research and Development Group, Khon Kaen University for funding this project.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.aanat.2016.04.029>.

References

Abrous, D.N., Koehl, M., Le Moal, M., 2005. Adult neurogenesis: from precursors to network and physiology. *Physiol. Rev.* 85, 523–569.
Bekinschtein, P., Kent, B.A., Oomen, C.A., Clemenson, G.D., Gage, F.H., Saksida, L.M., Bussey, T.J., 2014. Brain-derived neurotrophic factor interacts with adult-born

immature cells in the dentate gyrus during consolidation of overlapping memories. *Hippocampus* 24, 905–911.
Buckley, P.F., 2008. Update on the treatment and management of schizophrenia and bipolar disorder. *CNS Spectrums* 13, 1–10; quiz 11–12.
Cameron, H.A., McKay, R.D.G., 2001. Adult neurogenesis produces a large pool of new granule cells in the dentate gyrus. *J. Comp. Neurol.* 435, 406–417.
Conner, J.M., Lauterborn, J.C., Yan, Q., Gall, C.M., Varon, S., 1997. Distribution of brain-derived neurotrophic factor (BDNF) protein and mRNA in the normal adult rat CNS: evidence for anterograde axonal transport. *J. Neurosci.* 17, 2295–2313.
Couillard-Despres, S., Winner, B., Schauback, S., Aigner, R., Vroemen, M., Weidner, N., Bogdahn, U., Winkler, J., Kuhn, H.G., Aigner, L., 2005. Doublecortin expression levels in adult brain reflect neurogenesis. *Eur. J. Neurosci.* 21, 1–14.
Dix, S.L., Aggleton, J.P., 1999. Extending the spontaneous preference test of recognition: evidence of object-location and object-context recognition. *Behav. Brain Res.* 99, 191–200.
Duman, R.S., Nakagawa, S., Malberg, J., 2001. Regulation of adult neurogenesis by antidepressant treatment. *Neuropsychopharmacology* 25, 836–844.
Ehninger, D., Kempermann, G., 2008. Neurogenesis in the adult hippocampus. *Cell Tissue Res.* 331, 243–250.
Eriksson, P.S., Perfilieva, E., Bjork-Eriksson, T., Alborn, A.M., Nordborg, C., Peterson, D.A., Gage, F.H., 1998. Neurogenesis in the adult human hippocampus. *Nat. Med.* 4, 1313–1317.
Hawiset, T., Muchimapura, S., Wattanathorn, J., Sripanidkulchai, B., 2011. Screening neuropharmacological activities of *Kaempferia parviflora* (Krachai Dam) in healthy adult male rats. *Am. J. Appl. Sci.* 8, 695–702.
Hommet, C., Mondon, K., de Toffol, B., Constans, T., 2007. Reversible cognitive and neurological symptoms during valproic acid therapy. *J. Am. Geriatrics Soc.* 55, 628.
Hsieh, J., Nakashima, K., Kuwabara, T., Mejia, E., Gage, F.H., 2004. Histone deacetylase inhibition-mediated neuronal differentiation of multipotent adult neural progenitor cells. *Proc. Natl. Acad. Sci. U.S.A.* 101, 16659–16664.
Kempermann, G., 2006. *Adult Neurogenesis*. Oxford University Press, New York.
Kodama, M., Fujioka, T., Duman, R.S., 2004. Chronic olanzapine or fluoxetine administration increases cell proliferation in hippocampus and prefrontal cortex of adult rat. *Biol. Psychiatry* 56, 570–580.
Kostrouchova, M., Kostrouch, Z., Kostrouchova, M., 2007. Valproic acid, a molecular lead to multiple regulatory pathways. *Folia Biol.* 53, 37–49.
Kwan, P., Sills, G.J., Brodie, M.J., 2001. The mechanisms of action of commonly used antiepileptic drugs. *Pharmacol. Ther.* 90, 21–34.
Linnarsson, S., Willson, C.A., Ernfors, P., 2000. Cell death in regenerating populations of neurons in BDNF mutant mice. *Brain Res.* 75, 61–69.
Lyons, L., ElBeltagy, M., Bennett, G., Wigmore, P., 2012. Fluoxetine counteracts the cognitive and cellular effects of 5-fluorouracil in the rat hippocampus by a mechanism of prevention rather than recovery. *PLoS ONE* 7, e30010.
Mayhew, T.M., Burton, G.J., 1988. Methodological problems in placental morphometry: apologetics for the use of stereology based on sound sampling practice. *Placenta* 9, 565–581.
Mekjaruskul, C., Jay, M., Sripanidkulchai, B., 2012. Modulatory effects of *Kaempferia parviflora* extract on mouse hepatic cytochrome P450 enzymes. *J. Ethnopharmacol.* 141, 831–839.
Mukuda, T., Koyama, Y., Hamasaki, S., Kaidoh, T., Furukawa, Y., 2014. Systemic angiotensin II and exercise-induced neurogenesis in adult rat hippocampus. *Brain Res.* 1588, 92–103.
Mula, M., Trimble, M.R., 2009. Antiepileptic drug-induced cognitive adverse effects: potential mechanisms and contributing factors. *CNS Drugs* 23, 121–137.
Nibuya, M., Morinobu, S., Duman, R.S., 1995. Regulation of BDNF and trkB mRNA in rat brain by chronic electroconvulsive seizure and antidepressant drug treatments. *J. Neurosci.* 15, 7539–7547.
Nissinen, J., Pitkanen, A., 2007. Effect of antiepileptic drugs on spontaneous seizures in epileptic rats. *Epilepsy Res.* 73, 181–191.
Ramirez-Amaya, V., Marrone, D.F., Gage, F.H., Worley, P.F., Barnes, C.A., 2006. Integration of new neurons into functional neural networks. *J. Neurosci.* 26, 12237–12241.
Sairanen, M., Lucas, G., Ernfors, P., Castren, M., Castren, E., 2005. Brain-derived neurotrophic factor and antidepressant drugs have different but coordinated effects on neuronal turnover, proliferation, and survival in the adult dentate gyrus. *J. Neurosci.* 25, 1089–1094.
Santarelli, L., Saxe, M., Gross, C., Surget, A., Battaglia, F., Dulawa, S., Weisstaub, N., Lee, J., Duman, R., Arancio, O., Belzung, C., Hen, R., 2003. Requirement of hippocampal neurogenesis for the behavioral effects of antidepressants. *Science* 301, 805–809.
Shirayama, Y., Chen, A.C., Nakagawa, S., Russell, D.S., Duman, R.S., 2002. Brain-derived neurotrophic factor produces antidepressant effects in behavioral models of depression. *J. Neurosci.* 22, 3251–3261.
Spencer, J.P., 2009. Flavonoids and brain health: multiple effects underpinned by common mechanisms. *Genes Nutr.* 4, 243–250.
Surget, A., Saxe, M., Leman, S., Ibarguen-Vargas, Y., Chalon, S., Griebel, G., Hen, R., Belzung, C., 2008. Drug-dependent requirement of hippocampal neurogenesis in a model of depression and of antidepressant reversal. *Biol. Psychiatry* 64, 293–301.
Umka, J., Mustafa, S., ElBeltagy, M., Thorpe, A., Latif, L., Bennett, G., Wigmore, P.M., 2010. Valproic acid reduces spatial working memory and cell proliferation in the hippocampus. *Neuroscience* 166, 15–22.
Wang, D.S., Iwata, N., Hama, E., Saido, T.C., Dickson, D.W., 2003. Oxidized neprilysin in aging and Alzheimer's disease brains. *Biochem. Biophys. Res. Commun.* 310, 236–241.

- Wattanathorn, J., Pangpookiew, P., Sripanidkulchai, K., Muchimapura, S., Sripanidkulchai, B., 2007. Evaluation of the anxiolytic and antidepressant effects of alcoholic extract of *Kaempferia parviflora* in aged rats. *Am. J. Agric. Biol. Sci.* 2 (2), 94–98.
- Wetmore, C., Ernfors, P., Persson, H., Olson, L., 1990. Localization of brain-derived neurotrophic factor mRNA to neurons in the brain by in situ hybridization. *Exp. Neurol.* 109, 141–152.
- Winocur, G., Wojtowicz, J.M., Huang, J., Tannock, I.F., 2014. Physical exercise prevents suppression of hippocampal neurogenesis and reduces cognitive impairment in chemotherapy-treated rats. *Psychopharmacology* 231, 2311–2320.
- Zhang, Z., Yan, R., Zhang, Q., Li, J., Kang, X., Wang, H., Huan, L., Zhang, L., Li, F., Yang, S., Zhang, J., Ren, X., Yang, X., 2014. *Hes1*, a Notch signaling downstream target, regulates adult hippocampal neurogenesis following traumatic brain injury. *Brain Res.* 1583, 65–78.
- Zhao, C., Teng, E.M., Summers Jr., R.G., Ming, G.L., Gage, F.H., 2006. Distinct morphological stages of dentate granule neuron maturation in the adult mouse hippocampus. *J. Neurosci.* 26, 3–11.